From Luminal Kinematics to a Universal Causal Information Cone: Operationalizing Spacetime Geometry, Quantum Speed Limits, and Multi-Messenger Tests

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Abstract

For over a century, the invariant speed parameter of relativistic spacetime, $c \approx 2.99792458 \times 10^8$ m/s, has carried the historical nomenclature "the speed of light in vacuum." While historically natural following Maxwellian electrodynamics, this convention conflates a universal geometric property of spacetime with the excitation velocity of a specific $U(1)$ gauge field. In this work, we present a rigorous theoretical investigation that reformulates the kinematic invariant as the *universal causal speed of information* ($c_{\mathcal{I}}$), delineating established theorems, conceptual interpretations, and testable physical hypotheses. We establish: (i) **Axiomatic Foundations**: Following the Ignatowsky-Frank-Rothe-Lévy-Leblond lineage, Lorentz transformations and the Minkowski metric follow from isotropy, homogeneity, and group closure without postulating electromagnetism. Decoupling the metric from field excitations clarifies that if the photon possessed a non-zero Proca mass ($m_\gamma \neq 0$), physical light would be dispersive ($v_\gamma(\omega) < c_{\mathcal{I}}$) while the causal geometry of spacetime remains invariant. (ii) **Wave Dynamics and Quantum Speed Limits**: Via Kramers-Kronig analyticity and Sommerfeld-Brillouin contours, we reaffirm that front velocities strictly satisfy $v_{\text{front}} = c_{\mathcal{I}}$ despite superluminal group or phase velocities ($v_g, v_p > c_{\mathcal{I}}$). In subluminal media (Electromagnetically Induced Transparency), we show that group delays reflect steep polaritonic dispersion; we critically delineate this dispersive slowdown from isolated-system Margolus-Levitin quantum speed limits. (iii) **Proper Time as an Operational State-Evolution Constraint**: We show that the 4-velocity norm $u^\mu u_\mu = -c_{\mathcal{I}}^2$ operationally constrains the rate of quantum state distinguishability (measured via the Fubini-Study / Bures metric and quantum speed limits) along a worldline ($ds_{\text{FS}}/dt \leq 2\Delta E/(\hbar\gamma)$), rather than positing an ad-hoc "bandwidth fluid." (iv) **Hierarchy of Velocities**: We formulate a five-tier taxonomy and operational projection hierarchy ($c_{\text{geom}} \equiv c_{\mathcal{I}} \supset \{v_{\text{front}}, v_{\text{signal}}, v_{\text{LR}}, v_B\}$) rigorously distinguishing the invariant geometric causal speed c_{causal}, the field-theoretic front velocity v_{front}, effective Lieb-Robinson speeds v_{LR} in lattice systems, quantum scrambling velocities v_B, and usable signaling rates v_{signal}. We address the Renormalization Group (RG) conditions under which Lorentz-invariant cones emerge from microscopic lattice models. (v) **Entanglement and Spacetime**: We review the conditional derivation of Einstein's equations from entanglement equilibrium ($G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c_{\mathcal{I}}^4} T_{\mu\nu}$), maintaining an explicit Critical Audit Balance Sheet distinguishing conditionally derived structures from assumed geometric inputs. (vi) **The Universal Causal Cone Hypothesis**: We formulate the hypothesis $c_{\text{geom}} = c_{\text{EM}} = c_{\text{grav}} = c_{\text{matter}} = c_{\text{info}} \equiv c_{\mathcal{I}}$ as a falsifiable empirical proposition, evaluating current constraints from multi-messenger gravitational wave astronomy ($|c_{\text{grav}} - c_{\text{EM}}|/c_{\mathcal{I}} \leq 10^{-15}$ from GW170817), Proca mass bounds ($m_\gamma < 10^{-18}$ eV$/c_{\mathcal{I}}^2$), and Planck-scale modified dispersion relations.

Keywords: Foundations of Relativity, Causal Structure, Microcausality, Lieb-Robinson Bounds, Quantum Speed Limits, Holography, Multi-Messenger Astronomy.

1 Introduction: Decoupling Causal Geometry from Electromagnetism

Albert Einstein's 1905 formulation of Special Relativity [1] rested upon two postulates: the Principle of Relativity and the constancy of the speed of light in vacuum. Reconciling mechanics with Maxwell's wave equation,

$$\nabla^2 \mathbf{E} - \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}, \tag{1}$$

was a triumph of classical physics. Nevertheless, designating the fundamental invariant c as the "speed of light" anchored an incidental physical phenomenon—the propagation of transverse excitations in the abelian $U(1)$ gauge field—into the foundational geometry of spacetime.

This nomenclature has created persistent pedagogical and physical confusions. When light decelerates in dense matter or cold atomic vapors ($v_g \ll c$), students and researchers frequently confront apparent paradoxes regarding whether spacetime kinematics has altered. In anomalous dispersion regimes where group velocities exceed c ($v_g > c$) or become negative [8], claims of relativistic causality violations routinely re-emerge. Furthermore, from an axiomatic standpoint, if the photon should possess an infinitesimal rest mass $m_\gamma > 0$ as parameterized by de Broglie-Proca electrodynamics [27], electromagnetic waves would travel strictly below c with energy-dependent dispersion, yet the causal structure of spacetime would remain unaltered.

In modern theoretical physics, the emergence of spacetime geometry from quantum information—exemplified by the AdS/CFT correspondence [20], tensor networks [24], and thermodynamic derivations of gravity [17–19]—suggests that the causal structure of spacetime is fundamentally informational. The metric $g_{\mu\nu}$ defines the envelope of maximal causal influence: the universal speed of information $c_{\mathcal{I}}$.

1.1 Epistemic Framework: Three Tiers of Theoretical Status

To maintain rigorous scientific standards and avoid overclaiming, we explicitly partition the arguments in this study into three distinct epistemic tiers:

1. **Tier I: Mathematically Proven and Experimentally Established Physics.** This includes the group-theoretic derivation of Lorentz transformations without electrodynamics (the Ignatowsky lineage); the Sommerfeld-Brillouin front velocity analyticity stemming from Kramers-Kronig relations; relativistic microcausality in axiomatic QFT; Lieb-Robinson bounds in non-relativistic quantum spin systems; and the thermodynamic derivation of Einstein's equations from local entanglement equilibrium.

2. **Tier II: Conceptual and Operational Interpretations.** This includes viewing the invariant speed c as an informational causal horizon $c_{\mathcal{I}}$; interpreting proper time dilation $d\tau/dt = \sqrt{1 - v^2/c_{\mathcal{I}}^2}$ operationally as a constraint on the maximum rate of distinguishable quantum state transitions along a worldline; and synthesizing the taxonomy of physical velocities across quantum many-body systems and gravitational duals.

3. **Tier III: Falsifiable Hypotheses, Open Questions, and Theoretical Frontiers.** This includes the **Universal Causal Cone Hypothesis** ($c_{\mathrm{EM}} = c_{\mathrm{grav}} = c_{\mathrm{matter}} = c_{\mathcal{I}}$), subject to empirical multi-messenger tests; the Renormalization Group (RG) conditions governing the emergence of an exact continuum Lorentz light cone from discrete, non-relativistic microscopic information dynamics; and potential energy-dependent causal speeds $c_{\mathcal{I}}(E)$ in quantum gravity phenomenology.

By clearly isolating these three tiers, we ensure that established physics is respected, conceptual interpretations are defensible, and genuine open questions for theoretical and experimental research are identified.

2 Axiomatic Spacetime Foundations: Kinematics without Electromagnetism

2.1 The Ignatowsky-Frank-Rothe-Lévy-Leblond Lineage

It is a well-established mathematical theorem that Special Relativity does not logically require the existence of light. Beginning with W. von Ignatowsky (1910) [2], and further refined by Frank and Rothe (1911) [3], Lévy-Leblond (1976) [4], and Pelissetto and Testa (2015) [5], the Lorentz transformations can be deduced entirely from the following spacetime symmetries:

1. **Homogeneity of Spacetime:** Coordinate translations $(t, x) \to (t+\Delta t, x+\Delta x)$ preserve the form of physical laws, requiring transformation equations between inertial frames to be linear:
$$x' = A(v)x + B(v)t, \quad t' = C(v)x + D(v)t. \tag{2}$$

2. **Spatial Isotropy and Parity Invariance:** Spatial reflection $x \to -x$ must invert the sign of the relative boost velocity without altering physical laws.

3. **Group Closure and Reciprocity:** The set of boosts forms a one-parameter group $\mathcal{B}(v)$, satisfying $\mathcal{B}(v_2) \circ \mathcal{B}(v_1) = \mathcal{B}(v_3)$ with identity $\mathcal{B}(0) = \mathbb{I}$ and inverse $\mathcal{B}(-v) = \mathcal{B}(v)^{-1}$.

4. **Causality (Preclusion of Temporal Inversion):** Inertial transformations cannot reverse the temporal order of causally connectable events.

Enforcing these axioms uniquely determines the coordinate transformation up to a single universal constant α:

$$x' = \gamma(v)(x - vt), \quad t' = \gamma(v)\left(t - \alpha v x\right), \quad \gamma(v) = \frac{1}{\sqrt{1 - \alpha v^2}}. \tag{3}$$

Dimensional analysis dictates that $[\alpha] = [\text{velocity}]^{-2}$. Group theory admits precisely three distinct kinematic geometries:

- $\alpha < 0$**:** The symmetry group is the 4-dimensional Euclidean rotation group $SO(4)$. Spacetime possesses a positive-definite Riemannian metric signature $(+,+,+,+)$ lacking a hyperbolic light-cone structure; coordinate time behaves as a fourth spatial dimension without invariant causal ordering, which is ruled out empirically.

- $\alpha = 0$**:** The symmetry group is the Galilei group $\mathrm{SGal}(3,1)$, characterized by absolute Newtonian time and an infinite signal speed limit ($c_{\mathcal{I}} \to \infty$).

- $\alpha > 0$**:** The symmetry group is the Lorentz/Poincaré group $SO(3,1)$, possessing a pseudo-Riemannian hyperbolic structure with an invariant universal speed:
$$c_{\mathcal{I}} \equiv \frac{1}{\sqrt{\alpha}}. \tag{4}$$

Empirical reality selects $\alpha > 0$. Crucially, **no reference to Maxwell's equations, electric charges, or photons appears in this derivation**. The invariant velocity $c_{\mathcal{I}}$ is an intrinsic geometric feature of the pseudo-Riemannian manifold: it represents the maximal causal speed at which two spacetime events can exchange state alterations.

2.2 Decoupling Field Excitations: Maxwell vs. Proca Electrodynamics

The historical conflation of $c_{\mathcal{I}}$ with light arose because standard Maxwell electrodynamics assumes a strictly massless gauge boson ($m_\gamma = 0$):

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad \text{where } F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{5}$$

In Lorenz gauge $\partial_\mu A^\mu = 0$, the vacuum equation of motion is $\Box A^\mu = 0$, yielding the dispersion relation:

$$\omega^2 - c_{\mathcal{I}}^2 k^2 = 0 \implies v_{\text{phase}} = v_{\text{group}} = \frac{d\omega}{dk} = c_{\mathcal{I}}. \tag{6}$$

However, in de Broglie-Proca electrodynamics [27], the photon carries a non-zero invariant rest mass m_γ:

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\left(\frac{m_\gamma c_{\mathcal{I}}}{\hbar}\right)^2 A_\mu A^\mu. \tag{7}$$

The resulting field equation, $(\Box - m_\gamma^2 c_{\mathcal{I}}^2/\hbar^2)A^\mu = 0$, generates the relativistic dispersion relation:

$$\omega(k) = \sqrt{c_{\mathcal{I}}^2 k^2 + \frac{m_\gamma^2 c_{\mathcal{I}}^4}{\hbar^2}}. \tag{8}$$

The physical group velocity of light $v_\gamma(\omega)$ becomes frequency-dependent:

$$v_\gamma(\omega) = \frac{d\omega}{dk} = c_{\mathcal{I}}\sqrt{1 - \frac{m_\gamma^2 c_{\mathcal{I}}^4}{\hbar^2\omega^2}} < c_{\mathcal{I}}. \tag{9}$$

Under Proca electrodynamics, low-frequency radio waves propagate more slowly than high-energy gamma rays, and below the cutoff $\omega_{\text{cutoff}} = m_\gamma c_{\mathcal{I}}^2/\hbar$, propagation ceases. Yet, the Minkowski spacetime metric, Lorentz invariance, time dilation, and the universal causal horizon $c_{\mathcal{I}}$ remain entirely unaffected. This demonstrates that:

$$\boxed{c_{\mathcal{I}} \text{ is the invariant metric asymptote; } v_\gamma(\omega) \text{ is a dynamical field excitation velocity.}} \tag{10}$$

3 Wave Dynamics, Front Velocity, and Quantum State Evolution

3.1 Kramers-Kronig Analyticity and the Sommerfeld-Brillouin Front

A common source of confusion in wave physics is the observation of superluminal phase velocities ($v_p > c_{\mathcal{I}}$) or superluminal group velocities ($v_g > c_{\mathcal{I}}$) in media with anomalous dispersion or gain [8]. To establish that no physical signal can propagate faster than $c_{\mathcal{I}}$, consider a scalar field pulse $E(x,t)$ propagating through a dispersive linear medium with complex refractive index $n(\omega)$:

$$E(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{E}(0,\omega)\exp\left[i\left(\frac{\omega n(\omega)}{c_{\mathcal{I}}}x - \omega t\right)\right]d\omega. \tag{11}$$

Let the signal at $x = 0$ be initiated at $t = 0$, such that $E(0,t) = 0$ for all $t < 0$. By Titchmarsh's theorem, the Fourier transform $\tilde{E}(0,\omega)$ is analytic in the upper half of the complex frequency plane $\mathbb{C}^+ = \{\omega \in \mathbb{C} \mid \text{Im}(\omega) > 0\}$. The dielectric susceptibility $\chi(\omega) = n^2(\omega) - 1$ obeys the Kramers-Kronig causality relations. In the asymptotic high-frequency limit $|\omega| \to \infty$, physical charges cannot respond instantaneously to an oscillating driving field due to inertia, enforcing:

$$\lim_{|\omega|\to\infty} n(\omega) = 1. \tag{12}$$

Writing $\omega = \Omega + i\Xi$ with $\Xi > 0$, the exponent in the Fourier integral becomes:

$$\Phi(\omega) \equiv i\omega\left(\frac{n(\omega)x}{c_{\mathcal{I}}} - t\right) \xrightarrow{|\omega|\to\infty} i\Omega\left(\frac{x}{c_{\mathcal{I}}} - t\right) - \Xi\left(\frac{x}{c_{\mathcal{I}}} - t\right). \tag{13}$$

If $t < x/c_{\mathcal{I}}$, the coefficient $(x/c_{\mathcal{I}} - t)$ is strictly positive. Consequently, as $\Xi \to +\infty$, the term $-\Xi(x/c_{\mathcal{I}} - t) \to -\infty$, causing the integrand to vanish exponentially along any contour closed in the upper half-plane. By Jordan's lemma and Cauchy's residue theorem:

$$\oint_{\mathcal{C}} \tilde{E}(0,\omega)e^{\Phi(\omega)}d\omega = 0 \implies \boxed{E(x,t) \equiv 0 \quad \text{for all } t < \frac{x}{c_{\mathcal{I}}}.} \tag{14}$$

This classical result of Sommerfeld and Brillouin [6, 7] proves that the *front velocity* v_{front}, which governs the arrival of non-analytic signal discontinuities (information), is universally bounded by $c_{\mathcal{I}}$. Superluminal group velocities represent pulse-reshaping phenomena across the smooth body of an analytic packet, not superluminal information transfer.

3.2 Dark-State Polaritons and Slow Light: Dispersion vs. Quantum Speed Limits

At the opposite extreme, light can propagate at dramatically subluminal velocities ($v_g \ll c_{\mathcal{I}}$), reaching 17 m/s in ultracold sodium gases [9] or coming to a complete halt.

In Electromagnetically Induced Transparency (EIT) within a Λ-type three-level atomic ensemble driven by a classical control field Ω_c, the coupled probe-atom excitation forms a **dark-state polariton** [10]:

$$\hat{\Psi}(z,t) = \cos\theta\hat{\mathcal{E}}(z,t) - \sin\theta\sqrt{N}\hat{S}(z,t), \tag{15}$$

where $\hat{S}(z,t)$ is the collective atomic spin coherence operator, and the polariton mixing angle is defined by $\tan\theta = g\sqrt{N}/\Omega_c$, with g the single-photon atom-field coupling constant and N the atomic density. The resulting group velocity is:

$$v_g = c_{\mathcal{I}}\cos^2\theta = \frac{c_{\mathcal{I}}}{1 + \frac{g^2N}{\Omega_c^2}} = \frac{c_{\mathcal{I}}}{n_g}, \tag{16}$$

where $n_g = n + \omega\frac{dn}{d\omega} \gg 1$ is the group index.

Critical Distinction Regarding Quantum Speed Limits (QSL): It is important to clarify a conceptual nuance that is occasionally misstated in the literature. One cannot naively identify the macroscopic group delay $\Delta t_{\text{delay}} = L(n_g - 1)/c_{\mathcal{I}}$ as a sum of elementary Margolus-Levitin dwell times $\tau_{\text{dwell}} \geq \pi\hbar/(2g)$.

The Margolus-Levitin theorem [11] and the Mandelstam-Tamm bound [12] establish the minimum time required for an isolated quantum system under time-independent unitary evolution to evolve between mutually orthogonal states:

$$\tau_{\text{QSL}} \geq \max\left(\frac{\pi\hbar}{2\Delta E}, \frac{\pi\hbar}{2\langle E - E_0\rangle}\right). \tag{17}$$

In an EIT medium, the system is an open, driven, coupled atom-field system. The parameter g is a coupling frequency, not the total energy expectation value $\langle E - E_0\rangle$. The polaritonic slowdown is fundamentally an optical dispersion phenomenon: quantum destructive interference between excitation pathways eliminates resonant absorption while creating an extraordinarily steep refractive index gradient $dn/d\omega$. While the atomic coherence can store and retrieve quantum states, the slowdown is governed by polariton field kinematics rather than isolated-system QSL dwell times. The discontinuous wavefront continues to propagate at the invariant speed $c_{\mathcal{I}}$.

3.3 Operational Interpretation of Proper Time and Quantum State Evolution

The Lorentzian metric establishes the invariance of the timelike four-velocity norm:

$$u^\mu u_\mu = g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = -c_{\mathcal{I}}^2 \implies -c_{\mathcal{I}}^2\left(\frac{d\tau}{dt}\right)^2 + \left(\frac{d\mathbf{x}}{dt}\right)^2 = -c_{\mathcal{I}}^2. \tag{18}$$

Dividing by $-c_{\mathcal{I}}^2$ yields the well-known kinematic relation:

$$\left(\frac{d\tau}{dt}\right)^2 + \frac{v^2}{c_{\mathcal{I}}^2} = 1 \iff \frac{d\tau}{dt} = \sqrt{1-\frac{v^2}{c_{\mathcal{I}}^2}} = \frac{1}{\gamma(v)}. \tag{19}$$

Rather than interpreting this kinematic relation as an ontological "conservation of informational bandwidth fluid," its rigorous operational meaning connects spacetime geometry directly to quantum state distinguishability.

Consider a quantum clock characterized by an internal state $|\psi(\tau)\rangle$ evolving unitarily according to $i\hbar\frac{d}{d\tau}|\psi(\tau)\rangle = \hat{H}|\psi(\tau)\rangle$ in its comoving rest frame. The statistical distinguishability metric on projective Hilbert space is defined by the Fubini-Study / Bures line element $ds_{\text{FS}}^2 = 4(1-|\langle\psi(\tau)|\psi(\tau+d\tau)\rangle|^2) = \frac{4\Delta E^2}{\hbar^2}d\tau^2$, where $\Delta E^2 = \langle\hat{H}^2\rangle - \langle\hat{H}\rangle^2$ represents the energy variance, directly proportional to the Quantum Fisher Information $\mathcal{F}_Q(\tau) = 4\Delta E^2/\hbar^2$.

With respect to coordinate time t, the rate at which the clock traverses distinguishable quantum states along its worldline is constrained by:

$$\frac{ds_{\text{FS}}}{dt} = \frac{ds_{\text{FS}}}{d\tau}\frac{d\tau}{dt} = \frac{2\Delta E}{\hbar}\sqrt{1-\frac{v^2}{c_{\mathcal{I}}^2}} = \frac{2\Delta E}{\hbar\gamma(v)}. \tag{20}$$

Equivalently, applying the Mandelstam-Tamm [12] and Margolus-Levitin [11] Quantum Speed Limits (QSL), the rate of generating mutually orthogonal (fully distinguishable) states is bounded by:

$$\frac{dN_\perp}{dt} = \frac{dN_\perp}{d\tau}\frac{d\tau}{dt} \leq \min\left(\frac{2\Delta E}{\pi\hbar}, \frac{2\langle E-E_0\rangle}{\pi\hbar}\right)\frac{1}{\gamma(v)}. \tag{21}$$

As coordinate velocity approaches the causal limit ($v \to c_{\mathcal{I}}$), the rate of dynamical state distinguishability with respect to external coordinate time vanishes identically ($\lim_{v\to c_{\mathcal{I}}} ds_{\text{FS}}/dt = 0$). This establishes proper time dilation not as a metaphysical redistribution of an unphysical "information fluid," but as a precise geometric constraint on the physical rate of quantum state distinguishability and information processing along worldlines.

4 Quantum Information, Microcausality, and Non-Local Bounds

4.1 Microcausality and No-Signaling in Axiomatic QFT

In relativistic quantum field theory, the requirement that spacelike-separated events cannot causally influence each other is encoded in the **Microcausality Postulate**:

$$[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)] = 0 \quad \text{for all } (x-y)^2 = c_{\mathcal{I}}^2(t_x-t_y)^2 - (\mathbf{x}-\mathbf{y})^2 < 0. \tag{22}$$

Let Alice apply a local unitary transformation $\hat{U}_A = e^{-i\lambda\hat{\mathcal{O}}_A(x)}$ at event x. The expectation value of Bob's measurement of an observable $\hat{\mathcal{O}}_B(y)$ at spacelike separation y evaluates via the Baker-Campbell-Hausdorff formula:

$$\langle\hat{\mathcal{O}}_B(y)\rangle' = \text{Tr}\left(\rho\,\hat{U}_A^\dagger\hat{\mathcal{O}}_B(y)\hat{U}_A\right) = \langle\hat{\mathcal{O}}_B(y)\rangle + i\lambda\langle[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)]\rangle + \mathcal{O}(\lambda^2). \tag{23}$$

Because microcausality enforces $[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)] \equiv 0$ strictly for $(x-y)^2 < 0$, Bob's reduced density matrix is exactly invariant under Alice's local choice:

$$\rho_B = \mathrm{Tr}_A(\rho) \implies \frac{\partial \rho_B}{\partial \lambda} = 0 \implies \mathcal{C}_{A\to B} = 0 \quad \text{bits/s.} \tag{24}$$

Hence, quantum entanglement and projective measurement cannot transmit usable classical information across spacelike intervals, ensuring complete harmony between quantum non-locality and relativistic causality.

4.2 Information Causality as a Constraint on Non-Local Correlations

A foundational question in quantum information is why nature respects the Tsirelson bound for the Clauser-Horne-Shimony-Holt (CHSH) inequality ($S \leq 2\sqrt{2} \approx 2.828$), rather than permitting maximal non-signaling correlations such as Popescu-Rohrlich (PR) boxes ($S = 4$) [14].

Pawłowski et al. (2009) [15] demonstrated that the Tsirelson bound can be derived from the principle of **Information Causality**:

> *If Alice transmits an m-bit classical message to Bob, the total mutual information Bob can gain about Alice's database $\{a_k\}_{k=0}^{N-1}$ satisfies:*
>
> $$\sum_{k=0}^{N-1} I(a_k : b_k) \leq m. \tag{25}$$

In PR-box worlds, Bob could access an arbitrary number of Alice's bits while consuming only a single classical bit ($m = 1$), violating Information Causality. While Information Causality provides a compelling operational constraint that recovers the Tsirelson bound for bipartite settings, subsequent studies [16] have highlighted that whether Information Causality alone uniquely isolates the complete set of quantum correlations ($\mathcal{Q}$) across general multipartite scenarios remains an open research question, sensitive to the choice of entropic measures. Thus, Information Causality is best characterized as a *powerful necessary constraint* on physically admissible non-local correlations rather than an exhaustive axiomatic reconstruction of all quantum theory.

4.3 Quantum Teleportation and the Classical Channel Constraint

Quantum state teleportation encapsulates the necessity of the classical causal channel. While Alice's Bell-state measurement instantaneously projects Bob's entangled subsystem into one of four orthogonal unitary states $\{\sigma_k|\psi\rangle\}_{k=0}^{3}$, Bob's unconditioned state remains maximally mixed:

$$\rho_B = \frac{1}{4}\sum_{k=0}^{3} \sigma_k|\psi\rangle\langle\psi|\sigma_k^\dagger = \frac{1}{2}\mathbb{I}. \tag{26}$$

To complete the reconstruction, Bob must receive two classical bits transmitted via a physical channel at a speed $v \leq c_{\mathcal{I}}$. Until those bits arrive, the mutual information $I(\psi : \rho_B)$ is identically zero. Teleportation does not bypass relativistic causality; it strictly adheres to the speed of information $c_{\mathcal{I}}$.

5 A Unified Five-Tier Hierarchy of Physical Velocities

To prevent category errors and resolve conflations between different velocity definitions, we introduce a structured five-tier taxonomy of physical propagation velocities (Table 1).

Velocity Tier	Notation	Mathematical Definition	Physical Operational Domain
I. Geometric / Metric Causal Speed	$c_{\text{causal}} \equiv c_{\mathcal{I}}$	$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = 0$	Invariant asymptotic light-cone generator of pseudo-Riemannian spacetime.
II. Field-Theoretic Front Velocity	v_{front}	$\lim_{\omega\to\infty} \frac{\omega}{\text{Re}[k(\omega)]} = c_{\mathcal{I}}$	Speed of non-analytic signal discontinuities; strictly bounded by $c_{\mathcal{I}}$.
III. Many-Body Lattice Speed	v_{LR}	$\|[A(t), B]\| \leq Ce^{-\mu(d - v_{\text{LR}}t)}$	Effective quasi-causal cone in non-relativistic spin lattices ($v_{\text{LR}} \propto Ja/\hbar$).
IV. Quantum Chaos / Scrambling	v_B	$\langle [W(t,x), V(0)]^2 \rangle \sim e^{\lambda_L(t - \lvert x\rvert/v_B)}$	Operator spreading rate in strongly coupled CFTs ($v_B \leq c_{\mathcal{I}}$).
V. Usable Signal Velocity	v_{signal}	$\mathcal{C}_{A\to B}(t) > 0 \implies x/t \leq v_{\text{signal}}$	Maximal rate of message bit transfer through physical media ($v_{\text{signal}} \leq c_{\mathcal{I}}$).

Table 1: The Five-Tier Taxonomy of Physical Velocities across spacetime geometry, quantum many-body dynamics, and field theory.

This taxonomy clarifies that v_{front}, v_{signal}, v_{LR}, and v_B are not ontologically identical to $c_{\mathcal{I}}$; rather, they represent *distinct physical manifestations and operational projections of a single underlying geometric causal cone* under specific dynamical, material, and thermodynamic constraints:

$$c_{\text{geom}} \equiv c_{\mathcal{I}} \xrightarrow{\text{Projections}} \begin{cases} v_{\text{front}} \leq c_{\mathcal{I}} & \text{(High-frequency wave front)} \\ v_{\text{signal}} \leq c_{\mathcal{I}} & \text{(Shannon signaling in media)} \\ v_{\text{LR}} & \text{(Lattice quasi-cone bound)} \\ v_B \leq c_{\mathcal{I}} & \text{(Many-body scrambling)} \end{cases} \tag{27}$$

By organizing physical velocities according to this projection hierarchy, apparent paradoxes—such as anomalous superluminal group delays ($v_g > c_{\mathcal{I}}$) or subluminal polaritonic slowdowns ($v_g \ll c_{\mathcal{I}}$)—are systematically recognized as effective dynamical phenomena that leave the invariant geometric causal cone $c_{\mathcal{I}}$ intact.

6 Emergent Spacetime from Quantum Information

6.1 Lieb-Robinson Bounds: Effective Light Cones in Lattice Systems

In non-relativistic quantum mechanics, the Schrödinger operator $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi$ exhibits infinite propagation speed. However, in 1972, E. H. Lieb and D. W. Robinson [13] proved that for any quantum lattice system with short-range interactions, an effective speed limit emerges dynamically.

For local observables A and B supported on disjoint lattice subsets $X, Y \subset \Lambda$ separated by distance $d(X, Y)$:

$$\|[A(t), B(0)]\| \leq C\|A\|\,\|B\| \exp\left[-\mu\left(d(X,Y) - v_{\text{LR}}|t|\right)\right], \tag{28}$$

where the Lieb-Robinson velocity is bounded by $v_{\text{LR}} \leq \frac{2C\kappa}{\mu}e^1$, with κ characterizing the interaction decay scale.

Contrast with Relativistic Microcausality: It is crucial to emphasize the mathematical difference between Lieb-Robinson causality and relativistic microcausality:

- Relativistic QFT enforces **strict null causality**: outside the light-cone ($(x-y)^2 < 0$), commutators vanish identically ($[\mathcal{O}_A(x), \mathcal{O}_B(y)] \equiv 0$).
- The Lieb-Robinson bound defines an **effective quasi-causal cone**: outside the cone ($d > v_{\mathrm{LR}}|t|$), commutators decay exponentially, but are not strictly zero.

Furthermore, v_{LR} is a non-universal dynamical quantity determined by microscopic coupling parameters (typically $v_{\mathrm{LR}} \sim 10^3$ to 10^5 m/s in condensed matter systems), which is generally distinct from $c_{\mathcal{I}}$.

6.2 Renormalization Group Emergence of Lorentz Invariance

The profound theoretical question highlighted by this distinction is:

> *Under what Renormalization Group (RG) conditions does an exact, Lorentz-invariant continuum light cone emerge from microscopic quantum systems governed by Lieb-Robinson bounds?*

In quantum many-body systems near a Quantum Critical Point (QCP) characterized by a dynamic critical exponent $z = 1$, the low-energy effective action acquires space-time scaling isotropy:

$$\mathcal{S}_{\mathrm{eff}} = \int d^d x\, dt \left[\frac{1}{2}(\partial_t \phi)^2 - \frac{v_{\mathrm{eff}}^2}{2}(\nabla \phi)^2 - V(\phi) \right]. \tag{29}$$

In this infrared (IR) limit, the effective velocity $v_{\mathrm{eff}} \sim v_{\mathrm{LR}}$ plays the role of an emergent speed of light, and irrelevant Lorentz-violating operators flow to zero under RG coarse-graining:

$$\mathcal{H}_{\mathrm{micro}} \xrightarrow{\text{RG Coarse-Graining}} \mathcal{S}_{\mathrm{CFT}} \quad (z = 1 \implies SO(d,1) \text{ Lorentz symmetry}). \tag{30}$$

This provides a concrete theoretical mechanism by which a Lorentz-invariant causal structure can emerge in suitable microscopic systems at a $z = 1$ quantum critical point. We emphasize, however, that this represents a specific dynamical scenario rather than a universal emergence theorem; rigorously identifying the necessary and sufficient conditions under which generic non-relativistic lattice information networks flow to continuous pseudo-Riemannian geometries remains an active open research program.

6.3 Holographic Scrambling and the Butterfly Velocity

In holographic duality (AdS/CFT), strongly coupled conformal field theories on the boundary are dual to gravitational bulk spacetimes [20, 21]. The spreading of local perturbations in thermal states is measured by Out-of-Time-Ordered Correlators (OTOCs):

$$C(x,t) = -\langle [W(x,t), V(0)]^2 \rangle_\beta \sim \frac{1}{N^2} \exp\left[\lambda_L \left(t - \frac{|x|}{v_B} \right) \right], \tag{31}$$

where $\lambda_L \leq 2\pi k_B T/\hbar$ saturates the Maldacena-Shenker-Stanford chaos bound, and v_B is the **butterfly velocity**.

For Einstein gravity duals in AdS_{d+1}, the butterfly velocity takes the analytic form [22]:

$$v_B = \sqrt{\frac{d}{2(d-1)}} c_{\mathcal{I}} \leq c_{\mathcal{I}}. \tag{32}$$

It is essential to note that v_B is not the fundamental speed of light, but the rate of quantum chaotic scrambling across boundary degrees of freedom. In theories with higher-derivative curvature corrections (such as Gauss-Bonnet gravity) or anisotropic backgrounds, v_B departs from this specific formula while remaining bounded by the geometric causal speed ($v_B \leq c_{\mathcal{I}}$) under the relativistic causal assumptions considered here.

6.4 Entanglement Equilibrium and the Critical Audit Balance Sheet

Following Jacobson (1995, 2016) [17,18] and Faulkner et al. (2014) [19], Einstein's gravitational field equations can be recovered conditionally by imposing the **Entanglement Equilibrium Hypothesis** on small geodesic causal diamonds $\mathcal{D}(p, R)$ in a $D = 4$ pseudo-Riemannian manifold.

Let p be an arbitrary spacetime point and u^μ an arbitrary unit timelike vector ($u^\mu u_\mu = -c_{\mathcal{I}}^2$). In Riemann normal coordinates with $x^0 = c_{\mathcal{I}} t$, consider a spatial ball Σ_0 of radius R on a local maximal slice. The conformal Killing vector field inside the diamond generates modular flow with conformal factor $\xi^t(r) = \frac{R^2 - r^2}{2Rc_{\mathcal{I}}}$.

For a conformal field theory or the short-distance vacuum limit of general quantum fields, the modular Hamiltonian associated with the vacuum reduced density matrix ρ_0 takes the form:

$$\hat{K} = \frac{2\pi}{\hbar} \int_{\Sigma_0} \xi^t(r) \hat{T}_{00} d^3x, \tag{33}$$

where $\hat{T}_{00}$ is the physical energy density operator ($[\hat{T}_{00}] = \mathrm{J/m^3}$). Under a matter perturbation, the first law of entanglement entropy $\delta S_{\mathrm{matter}} = \delta \langle \hat{K} \rangle$ gives:

$$\delta S_{\mathrm{matter}} = \frac{2\pi}{\hbar} T_{00}^{\mathrm{phys}} \int_0^R \frac{R^2 - r^2}{2Rc_{\mathcal{I}}} (4\pi r^2 dr) = \frac{8\pi^2 R^4}{15\hbar c_{\mathcal{I}}} T_{00}^{\mathrm{phys}}. \tag{34}$$

Simultaneously, spacetime curvature induces a geometric area deficit of the geodesic boundary sphere $\partial\Sigma_0$. At fixed spatial 3-volume ($\delta V = 0$), the area deficit evaluated from the Raychaudhuri equation is:

$$\delta A = -\frac{4\pi R^4}{15} G_{00}^{\mathrm{geom}}, \tag{35}$$

where $[G_{00}^{\mathrm{geom}}] = 1/\mathrm{m^2}$. Postulating that the unperturbed UV vacuum entanglement entropy is proportional to the horizon area with Bekenstein-Hawking coefficient $\eta = \frac{c_{\mathcal{I}}^3}{4G\hbar}$ ($[\eta] = 1/\mathrm{m^2}$), the geometric entanglement change is:

$$\delta S_{\mathrm{UV}} = \eta \delta A = -\frac{\pi c_{\mathcal{I}}^3 R^4}{15 G\hbar} G_{00}^{\mathrm{geom}}. \tag{36}$$

Imposing the Entanglement Equilibrium Hypothesis ($\delta S_{\mathrm{total}} = \delta S_{\mathrm{UV}} + \delta S_{\mathrm{matter}} = 0$ for all local diamonds at fixed volume) yields:

$$-\frac{\pi c_{\mathcal{I}}^3 R^4}{15 G\hbar} G_{00}^{\mathrm{geom}} + \frac{8\pi^2 R^4}{15\hbar c_{\mathcal{I}}} T_{00}^{\mathrm{phys}} = 0 \implies G_{00}^{\mathrm{geom}} = \frac{8\pi G}{c_{\mathcal{I}}^4} T_{00}^{\mathrm{phys}}. \tag{37}$$

Because this scalar projection holds for all unit timelike vectors u^μ, tensor multilinearity and local conservation $\nabla^\mu T_{\mu\nu} = 0$ together with the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ require the inclusion of a constant of integration $\Lambda g_{\mu\nu}$ associated with the volume constraint:

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c_{\mathcal{I}}^4} T_{\mu\nu}.} \tag{38}$$

To avoid claiming that gravity is derived "from pure information alone," Table 2 provides an explicit **Critical Audit Balance Sheet** distinguishing conditionally derived features from assumed empirical inputs.

Component	Methodological Status	Physical Description / Role
Einstein Tensor ($G_{\mu\nu}$)	**Conditionally Derived**	Generated by multilinear contraction across all time-like observer 4-velocities u^μ.
Stress-Energy Coupling ($T_{\mu\nu}$)	**Conditionally Derived**	Dictated by modular Hamiltonian flow generating local boost thermalization.
Coupling Ratio ($\kappa = 8\pi G/c_{\mathcal{I}}^4$)	**Conditionally Derived**	Emerges from thermodynamic area-to-matter entropy balance ($\eta/c_{\mathcal{I}}$).
Energy-Momentum Conservation	**Conditionally Derived**	Guaranteed by the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$.
Newton's Constant G	**ASSUMED INPUT**	Enters via the empirical Bekenstein-Hawking coefficient $\eta = c_{\mathcal{I}}^3/4G\hbar$.
Smooth Lorentzian Manifold $(M, g_{\mu\nu})$	**ASSUMED INPUT**	Assumes a pre-existing 4D smooth continuum with Riemann normal coordinates.
Entanglement Equilibrium Axiom	**ASSUMED INPUT**	Variational stationarity $\delta S_{\text{total}} = 0$ is an operational thermodynamic postulate.

Table 2: Critical Audit Balance Sheet: Conditional derivation of Einstein's equations from the Entanglement Equilibrium Hypothesis.

6.5 Holographic Quantum Error Correction and Spacetime Fault Tolerance

Why does emergent semiclassical spacetime appear macroscopically smooth and stable against ultraviolet vacuum fluctuations? Following Almheiri, Dong, and Harlow (2015) [23] and the HaPPY tensor network code [24], semiclassical bulk states form a protected logical code subspace $\mathcal{H}_{\text{code}} \subset \mathcal{H}_{\text{phys}}$ satisfying the Knill-Laflamme condition:

$$P_{\text{code}}\, E_a^\dagger E_b\, P_{\text{code}} = C_{ab}\, P_{\text{code}}. \tag{39}$$

Within this framework, the holographic radial depth z corresponds to the quantum code distance $d(z)$, shielding bulk degrees of freedom against boundary erasures.

Furthermore, the Eastin-Knill theorem [25] proves that no quantum error-correcting code can implement a continuous universal set of logical gates purely through transversal (spatially local and instantaneous) operations. In any physically realistic, spatially distributed implementation governed by local Hamiltonian interactions, syndrome extraction, measurement, and active feedback operations are necessarily subject to the causal propagation constraints of the underlying hardware or field theory. While the Eastin-Knill theorem does not itself derive relativistic light cones, it reinforces the operational principle that active fault-tolerant stabilization across macroscopic scales cannot be executed instantaneously, but requires physical causal channels bounded by $c_{\mathcal{I}}$.

7 The Universal Causal Cone Hypothesis and Empirical Signatures

7.1 The Universal Causal Cone Hypothesis

Replacing c with $c_{\mathcal{I}}$ transitions from an interpretive relabeling to a testable physical theory when we distinguish the causal speeds across different fundamental sectors. Let:

- c_{geom} denote the null cone generator of the spacetime metric $g_{\mu\nu}$,
- c_{EM} denote the vacuum propagation speed of electromagnetic excitations,

- c_{grav} denote the propagation speed of gravitational tensor waves,
- c_{matter} denote the limiting velocity of fermionic matter fields, and
- c_{info} denote the supremum of causal information transmission.

We formulate the **Universal Causal Cone Hypothesis**:

$$\boxed{c_{\text{geom}} = c_{\text{EM}} = c_{\text{grav}} = c_{\text{matter}} = c_{\text{info}} \equiv c_{\mathcal{I}}.} \tag{40}$$

This hypothesis is not a mathematical tautology; it is a falsifiable physical proposition regarding the universality of Lorentz invariance across all quantum fields. In standard General Relativity coupled to the Standard Model, the equality of light and gravitational wave velocities is empirically incorporated via global and local Lorentz symmetry. Positing the Universal Causal Cone Hypothesis elevates this empirical coincidence to an axiomatic principle across all sectors: geometry, gauge fields, fermions, and information. If any sector possessed a distinct propagation speed ($\Delta c_i = c_i - c_{\mathcal{I}} \neq 0$), Lorentz symmetry would be broken or deformed, yielding distinct physical light cones for different field excitations.

7.2 Multi-Messenger Gravitational Wave Constraints

The most stringent empirical test of the equality $c_{\text{grav}} = c_{\text{EM}}$ was achieved through the simultaneous observation of the binary neutron star merger GW170817 and its electromagnetic counterpart GRB 170817A [26].

The signals traversed a cosmological distance of $D \approx 40$ Mpc ($\approx 1.3 \times 10^8$ light-years) with an observed arrival time difference of $\Delta t = 1.74 \pm 0.11$ s. Accounting for astrophysical emission delays, this measurement constrained the relative speed difference to:

$$-3 \times 10^{-15} \leq \frac{c_{\text{grav}} - c_{\text{EM}}}{c_{\mathcal{I}}} \leq +7 \times 10^{-16}. \tag{41}$$

This precision measurement demonstrates that gravitational and electromagnetic radiation share an identical propagation speed to within one part in 10^{15}, strongly constraining any sector-dependent deviations from the Universal Causal Cone Hypothesis.

7.3 Astrophysical Proca Photon Mass Limits

If the photon possesses a Proca mass m_γ, the propagation speed $v_\gamma(\nu)$ varies with frequency ν. For two signals emitted simultaneously from an astrophysical source at distance D, the differential arrival delay is:

$$\Delta t = \frac{D}{c_{\mathcal{I}}} \left(\frac{1}{\sqrt{1 - \frac{m_\gamma^2 c_{\mathcal{I}}^4}{h^2 \nu_1^2}}} - \frac{1}{\sqrt{1 - \frac{m_\gamma^2 c_{\mathcal{I}}^4}{h^2 \nu_2^2}}} \right) \approx \frac{D m_\gamma^2 c_{\mathcal{I}}^3}{2h^2} \left(\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right). \tag{42}$$

Using Fast Radio Bursts (FRBs) and pulsar timing arrays across radio frequencies (10 MHz – 10 GHz), current astrophysical bounds constrain the photon mass to:

$$m_\gamma < 10^{-18} \text{ eV}/c_{\mathcal{I}}^2 \approx 1.8 \times 10^{-54} \text{ kg}. \tag{43}$$

Should future ultra-low-frequency observations detect a non-zero photon mass, the physical speed of light v_γ would be dispersive, whereas $c_{\mathcal{I}}$ would remain the invariant causal asymptote.

7.4 Modified Dispersion Relations: Group Velocity Delay vs. Causal Cones

In candidate quantum gravity theories (such as Loop Quantum Gravity and D-brane string models), Planck-scale spacetime granularity induces Modified Dispersion Relations (MDR):

$$E^2 = p^2 c_{\mathcal{I}}^2 \left[1 \pm \xi \left(\frac{E}{E_{\mathrm{QG}}}\right)^n\right], \quad n \in \{1, 2\}. \tag{44}$$

This modifies the particle group velocity:

$$v_g(E) = \frac{\partial E}{\partial p} \approx c_{\mathcal{I}} \left[1 \mp \frac{n+1}{2} \xi \left(\frac{E}{E_{\mathrm{QG}}}\right)^n\right]. \tag{45}$$

Critical Delineation: Group Velocity Delay vs. Metric Causal Cone: It is crucial to distinguish between an energy-dependent group velocity $v_g(E)$ and an energy-dependent deformation of the spacetime metric causal cone c_{geom}. In effective field theories with higher-spatial-derivative operators, particle wavepackets experience energy-dependent dispersive delays ($v_g(E) \neq c_{\mathcal{I}}$), yet the Sommerfeld-Brillouin front velocity v_{front} and the underlying metric null cone remain strictly invariant at $c_{\mathcal{I}}$. Only if Planck-scale physics alters the asymptotic high-frequency limit $\lim_{\omega\to\infty} \omega/k$ or introduces non-local spacetime granularity (such as in Doubly Special Relativity or rainbow gravity) does the fundamental causal cone become energy-dependent ($c_{\mathcal{I}}(E)$).

Observations by the Fermi Gamma-ray Space Telescope of GRB 090510 ($z = 0.903$, containing a 31 GeV photon arriving within 0.829 s of the low-energy pulse onset) constrained linear Lorentz invariance violation ($n = 1$) [28]:

$$E_{\mathrm{QG},1} > 1.2\, M_{\mathrm{Planck}} \approx 1.5 \times 10^{19}\ \mathrm{GeV}. \tag{46}$$

This measurement places stringent empirical bounds on dispersive particle delays, confirming the energy-independence of luminal propagation up to the Planck scale and strongly constraining candidate deformations of the Universal Causal Cone.

8 Comparative Assessment: Foundational and Operational Gains

Table 3 summarizes the foundational and explanatory gains achieved by transitioning from the speed of light to the speed of information.

9 Conclusion and Future Research Directions

The conceptual shift from the "speed of light" (c) to the "speed of information" ($c_{\mathcal{I}}$) elevates relativity from an electrodynamic postulate to a universal property of spacetime causality and quantum information. By separating established theorems from operational interpretations and empirical hypotheses, this framework resolves historical paradoxes, clarifies the role of quantum speed limits, and establishes a coherent vocabulary for quantum gravity.

We propose three primary directions for future mathematical and empirical research:

1. **Renormalization Group Flows from Quantum Lattices to Lorentz Geometry:** Formulating explicit lattice Hamiltonians whose Lieb-Robinson velocity v_{LR} flows to an exact continuum light-cone generator $c_{\mathcal{I}}$ under non-perturbative RG techniques.

2. **Non-AdS Holographic Scrambling Bounds:** Extending butterfly velocity relations $v_B \leq c_{\mathcal{I}}$ to asymptotically flat and cosmological de Sitter horizons.

3. **Precision Multi-Messenger Limits on Causal Universality:** Utilizing next-generation gravitational wave detectors (LISA, Einstein Telescope) and ultra-high-energy neutrino observatories (IceCube-Gen2) to test $|c_{\mathrm{grav}} - c_\nu|/c_{\mathcal{I}}$ and $|c_{\mathrm{grav}} - c_{\mathrm{EM}}|/c_{\mathcal{I}}$ beyond 10^{-17}.

Dimension	Explanatory Gain	Theoretical & Empirical Impact
1. Axiomatic Kinematics	Eliminates Maxwellian Dependency	Decouples spacetime geometry from abelian gauge theory. Proves Lorentz invariance follows from symmetry and causality, preserving validity if $m_\gamma \neq 0$.
2. Wave Dynamics	Resolves Superluminal Paradoxes	Confirms that anomalous group delays ($v_g > c$) are envelope reshaping artifacts, while true information transfer is bounded by $v_{\text{front}} = c_{\mathcal{I}}$.
3. Quantum Clocks	Operational Proper Time	Interprets $d\tau/dt = 1/\gamma$ as a geometric constraint on the rate of distinguishable quantum state transitions bounded by QSLs, rather than an ad-hoc fluid.
4. Velocity Taxonomy	Resolves Category Conflations	Establishes a five-tier hierarchy cleanly separating c_{causal}, v_{front}, v_{LR}, v_B, and v_{signal}.
5. Emergent Spacetime	Aligns with Quantum Information	Connects Lieb-Robinson bounds, tensor network codes, and entanglement equilibrium to the emergence of macroscopic geometry.
6. Empirical Signatures	Falsifiable Physics	Frames multi-messenger bounds ($\lvert c_{\text{grav}} - c_{\text{EM}} \rvert \leq 10^{-15}$), Proca masses, and MDRs as tests of the Universal Causal Cone.

Table 3: Systematic evaluation of theoretical and empirical gains from the $c_{\mathcal{I}}$ framework.

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